



Orator

Code Review & Maintenance Report

Date: 2026-06-17

Repository: mickhatzikostas220-design/speech-analyzer2

Branch: claude/relaxed-sagan-reg96i

Commits in this pass: 5

This report documents a review-and-fix pass over the Orator web application (a Next.js 14 / TypeScript / Supabase app for fMRI-based neural speech analysis). The goal was to find and fix anything across the site that needed attention. Every change was verified with a full production build (`next build`) and a lint pass before being committed.

Headline result

- Fixed a production build-breaking bug — the app would not build for deployment before this pass.
- Established a working ESLint setup and fixed all lint errors.
- Unified the product brand name (removed leftover 'ACA' references in user-facing emails and the AI chat assistant).
- Added a missing favicon / app icon and richer page metadata.
- Removed a committed build-cache file from version control.

1. Build-breaking bug: module-level OpenAI client

Severity: HIGH | Files: lib/openai.ts, app/api/compare-report/route.ts

Problem

Both files created the OpenAI client at module scope (`new OpenAI({ ... })`). During `next build`, Next.js evaluates route modules while collecting page data; if `OPENAI_API_KEY` is not set at that moment the constructor throws and aborts the whole build with 'The OPENAI_API_KEY environment variable is missing or empty'. The build failed outright at `/api/compare-report`.

Fix

Moved client construction to lazy initialization so it runs at request time, not at import/build time. `lib/openai.ts` now uses a cached `getOpenAI()` helper; the `compare-report` route constructs the client inside the POST handler.

```
let _openai = null;
function getOpenAI() {
  if (!_openai) _openai = new OpenAI({ apiKey: process.env.OPENAI_API_KEY });
  return _openai;
}
```

Verification

`next build` now completes: 'Compiled successfully' and 'Generating static pages (21/21)'.

2. ESLint configuration + lint errors

Severity: MEDIUM | Files: .eslintrc.json (new), package.json, 5 page files

Problem

The project had no ESLint configuration, so `npm run lint` dropped into an interactive setup prompt and could not run non-interactively. Several pages also contained unescaped quotes/apostrophes in JSX (`react/no-unescaped-entities`).

Fix

- Added `.eslintrc.json` extending `next/core-web-vitals`.
- Installed the missing `@typescript-eslint` plugin so existing inline disable directives resolve, and turned off the noisy `no-explicit-any` rule for this codebase.
- Escaped quotes/apostrophes in the editor, history, signup, request-access and request-access/success pages.

Verification

`npm run lint` now runs clean — only two pre-existing `<img>` performance warnings remain, which do not fail the build.

3. Brand-name inconsistency: 'ACA' → 'Orator'

Severity: MEDIUM (user-facing) | Files: components/Navbar.tsx, lib/email.ts, app/api/analyses/[id]/chat/route.ts

Problem

The product is named 'Orator' in the README, page titles and auth screens, but several places still showed a prior name, 'ACA'. This was visible to end users: the top-nav logo, all three transactional emails (access-request notification, approval, rejection), and the identity of the in-app AI chat assistant.

Fix

- Navbar logo text: ACA → Orator.
- lib/email.ts: sender name, body copy and footers across all three emails.
- Chat assistant system prompt now reads 'You are Orator's speech analysis assistant'.

A repository-wide search confirms no 'ACA' references remain in app / lib / components.

4. Missing favicon and thin metadata

Severity: LOW | Files: app/icon.svg (new), app/layout.tsx

Problem

The site shipped no favicon / app icon (no public/ directory, no app/icon). Page metadata was minimal — only a title and description, with no metadataBase, social cards or robots policy.

Fix

- Added app/icon.svg — a gradient brand mark matching the in-app logo, auto-detected by Next.js.
- Expanded root metadata: metadataBase, a title template, OpenGraph + Twitter cards, and robots index:false / follow:false (appropriate because Orator is an invite-only, private app that should not be indexed).

5. Build cache committed to git

Severity: LOW | Files: .gitignore, tsconfig.tsbuildinfo (removed)

Problem

tsconfig.tsbuildinfo — a ~115 KB machine-specific TypeScript incremental build cache — was tracked in version control, causing noisy diffs and potential merge conflicts.

Fix

Added *.tsbuildinfo to .gitignore and removed the file from tracking (git rm --cached).

Areas reviewed and found healthy

The following were inspected and required no changes:

- Auth middleware (route protection / redirects).
- Upload flow (UploadZone), including timeout handling and file validation.
- Analysis detail page: polling, rename, delete, retry, export and streaming chat.
- Export route (.txt / .csv / .json) with CSV quote-escaping.
- History page: search + pagination.
- Compare page: metric overlay chart and prompt generator.
- Public access-request API: duplicate detection + validation.
- Other API clients (Supabase admin, Resend) already use lazy / in-handler init.

Suggested follow-ups (not done this pass)

- Dashboard / analysis pages poll every 2.5s while a job runs 10–20 min (hundreds of requests); consider backoff or realtime subscriptions.
- No API rate limiting (already noted in README known-limitations).
- CSV export could harden against spreadsheet formula injection (low risk — values are AI-generated).
- /api/compare-report (OpenAI-backed) appears unused; the Compare UI generates a copy/paste prompt instead. Wire up or remove.
- Replace `<img>` with `next/image` in the timeline editor to clear the remaining lint warnings.

Commit log

```
a3f53a0  Fix build failure from module-level OpenAI init; add ESLint config
f38c22b  Fix navbar brand name (ACA -> Orator) for consistency
805dde1  Add app icon and richer metadata (OG, Twitter, noindex)
f2ee11a  Replace stale 'ACA' brand name with 'Orator'
c1abcb5  Stop tracking tsconfig.tsbuildinfo build cache
```

All changes pushed to branch `claude/relaxed-sagan-reg96i`. Verified with a clean production build (21/21 routes) and a lint pass.